Fall 2021 Shopify Data Science Intern Challenge
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Question 1:

a. The average order value being calculated as $3145.13 is the **mean** of all order amounts in the data set. From a quick scatterplot of order amount vs. order ID (created in Excel), it is clear that there are several **outliers** in the data – most much larger than the majority of order values (i.e., bulk orders of $704,000 are common). These outliers greatly **distort the mean**.

b. A more appropriate metric for this data set is the **median**. The outliers that significantly distort the mean will have a much smaller effect on the median, making it a **more accurate measure of central tendency.**

c. Using Excel, with the formula:

```
=MEDIAN(D2:D5001)
```

The revised AOV, using the **median**, is **$284**.

Question 2:

a. SQL query:

```
SELECT COUNT (Orders.OrderID)
FROM Orders
WHERE ShipperID =
        (SELECT ShipperID
        FROM Shippers
         WHERE ShipperName = 'Speedy Express')
```

54 orders were shipped by Speedy Express, in total.

b. SQL query:

```
SELECT Employees.LastName
FROM Employees
WHERE Employees.EmployeeID =
        (SELECT EmployeeID
        FROM (
                SELECT EmployeeID, MAX (orderCount)
                FROM (
                SELECT Orders.EmployeeID, COUNT(Orders.OrderID) orderCount
                FROM Orders
                GROUP BY Orders.EmployeeID)))
```

The last name of the employee with the most orders is **Peacock.**

c. SQL query:

```
SELECT Products.ProductName, SUM(OrderDetails.Quantity)
FROM Orders
JOIN Products
        ON Products.ProductID = OrderDetails.ProductID
JOIN OrderDetails
        ON OrderDetails.OrderID = Orders.OrderID
JOIN Customers
        ON Orders.CustomerID=Customers.CustomerID
WHERE Customers.Country = 'Germany'
GROUP BY OrderDetails.ProductID
ORDER BY SUM(OrderDetails.Quantity) DESC
LIMIT 1
```

The product ordered the most by customers in Germany was **Boston Crab Meat**, with **160** orders.